

EduVision_DV — QA Checklist (Module 7: Testing & Validation)

Based on direct visual inspection of all 4 live dashboards (screenshots dated 2026-10-02). Items marked FAIL are confirmed defects with cited evidence — not assumptions.

#	Category	Check Item	Status	Severity	Evidence / Notes
SECTION 1 — Confirmed Defects (found by direct visual inspection)					
1	Filter Logic	Top 10 Universities ranking shows exactly 10 rows	FAIL	MAJOR	University Overview: 'Top 10 Universities by Global Ranking Score' renders 13 rows (Harvard through 2nd 'University of Califor..'). Top N filter is not limiting correctly.
2	Filter Logic	Top 5 by Research Productivity Index shows exactly 5 rows	FAIL	MAJOR	University Overview: renders 15 rows (Harvard through Albert Ludwig). Same systemic Top-N filter bug as item 1 — likely affects every Top-N chart across all 4 dashboards.
3	Field Formatting	Year filter displays as a plain year (2005), not a formatted number	FAIL	MODERATE	All 4 dashboards: Year filter shows '2,005' with a thousands separator. The Year field is still typed/formatted as a continuous number rather than a discrete year — same root cause flagged in earlier Tableau setup steps.
4	Chart Rendering	Bar length is proportional to its labeled value	FAIL	MAJOR	Research Analytics: 'Top 10 Publications' — Harvard (1,100.0) and Columbia (810.0), a 36% difference, render as near-identical bar lengths. The bar encoding does not match the data labels.
5	Aggregation Consistency	A single chart uses ONE consistent aggregation for both the mark size/length and its label	FAIL	MAJOR	Student Analytics: 'International Students % by University' — axis scale reaches 300-400 while data labels show 37-55%. Label likely uses AVG; bar height appears to use SUM across ~8 years of records for the same university.
6	Cross-Dashboard Consistency	The same KPI reads the same value on every dashboard it appears on	FAIL	MAJOR	International Student % = 15.9% on University Overview, but Student Analytics implies ~20.0% (Domestic Enrollment 14.7K vs Avg Enrollment 18.38K -> $3.68K \text{ intl} / 18.38K = 20.0\%$). Same underlying metric, two different answers.
7	Data Integrity	Different measures in a comparison table produce genuinely different values	FAIL	MAJOR	Country Comparison: ranking table's 'academi..' (academic_reputation_score) and 'global_r..' (global_ranking_score) rows are IDENTICAL in 4 of 5 country columns (Canada, Germany, Japan, UK, US) and differ only in the final row value. Strongly suggests a duplicated or miscalculated measure.

8	Labeling	Row/column headers show friendly names, not raw database field names	FAIL	MINOR	University Overview and Country Comparison tables display raw field names ('university_name', 'academi..', 'global_r..') as row headers instead of 'University' / 'Academic Reputation' / 'Global Ranking Score'.
9	Chart Legends	Every multi-series chart has a legend identifying each series	FAIL	MINOR	Country Comparison: 'Country Radar Comparison' shows two overlapping shapes (orange, small; blue-grey, larger) with no legend. Cannot tell which country or series each shape represents.
10	Null Handling	Charts with missing data show a visible, explained null indicator (not silently dropped)	VERIFY	MODERATE	Multiple charts show null-count badges: 14 nulls (ranking trend, x2), 195 nulls (impact-vs-productivity scatter), 19 nulls (productivity trend), 4 nulls (treemap), 1 null (diversity trend), 1 unknown (map). Badges ARE showing — confirm the underlying cause of each is understood and acceptable, not a pipeline defect.

SECTION 2 — KPI Calculation Validation (verify manually against source data)

11	KPI Accuracy	Global Ranking Score matches AVG(global_ranking_score) for the filtered selection	VERIFY	MAJOR	Spot-check 3 universities against university_final_dataset.csv directly
12	KPI Accuracy	Faculty Ratio displays as '1 : X' with X = AVG(faculty_student_ratio)	VERIFY	MODERATE	Confirm format string applied correctly, not accidentally inverted (1:18.4 vs 18.4:1)
13	KPI Accuracy	International Student % is the same calculation method on every dashboard it appears	FAIL	MAJOR	See defect #6 above — this is currently failing
14	KPI Accuracy	Research Productivity Index and Research Impact Score are NOT the same underlying calculation	VERIFY	MODERATE	Confirm these two KPIs use genuinely different formulas, not the same field relabeled
15	KPI Accuracy	Region totals in the donut chart sum to the Global Universities KPI (1,373)	VERIFY	MODERATE	University Overview: Europe 493 + Asia 335 + South America 43 + Africa 138/16(unclear) — verify exact segment values sum to 1,373
16	Ranking Logic	Rank #1 (Top Global Ranking KPI) matches the #1 row in the Top 10 Ranking table	VERIFY	MAJOR	KPI card shows 'California Institute of Technology' as Top Global Ranking, but the Top 10 table's #1 row is Harvard University — these should match and currently appear to disagree

SECTION 3 — Dashboard Interaction Testing

17	Navigation	Clicking each top-nav button (Research Analytics / Student Analytics / Country Comparison / University Overview) opens the correct dashboard	VERIFY	MAJOR	Test all 12 possible nav paths (4 dashboards x 3 destinations each)
18	Filters	Changing Year filter updates every chart on the current dashboard	VERIFY	MAJOR	Test with at least 3 different year values, confirm all charts respond
19	Filters	Changing Region filter narrows the Country filter's available options	VERIFY	MODERATE	Select a Region, confirm Country dropdown only shows countries in that region

20	Filters	Filter state persists correctly when switching between dashboards	VERIFY	MODERATE	Set Year=2015, switch dashboards, confirm Year filter still shows 2015
21	Cross-filtering	Clicking a mark on one chart filters related charts on the same dashboard	VERIFY	MAJOR	Click a university bar, confirm other charts narrow to that selection; click empty space to confirm reset
22	Tooltips	Hovering any mark shows accurate values matching the visible label	VERIFY	MINOR	Spot-check 5 marks across different chart types
23	Map Interaction	Map markers are clickable and show correct country/university on hover	VERIFY	MODERATE	International Students Map and Country map — test hover and click behavior

SECTION 4 — Educational Metrics Validation

24	Metric Validity	Faculty-to-Student Ratio values fall within a plausible real-world range (typically 1:5 to 1:30)	PASS	-	Observed range 1:12.3 to 1:21.4 by region — plausible
25	Metric Validity	International Student % values fall within 0-100% for every row	VERIFY	MODERATE	Confirm no values exceed 100% or go negative after the aggregation bug (defect #5) is fixed
26	Metric Validity	Research Productivity Index and Research Impact Score are both bounded 0-100 as documented	VERIFY	MINOR	Research Analytics scatter plot axes both go to 100 — confirm no values exceed that bound
27	Metric Validity	Avg Overall Score (16.8) is consistent with the 0-100 scale used elsewhere	VERIFY	MODERATE	16.8 looks low compared to other score KPIs in the 30-70 range on the same dashboard — confirm this isn't a scale mismatch

Legend: FAIL = confirmed defect, cited evidence · VERIFY = requires manual check against live data, not yet confirmed either way · PASS = checked and correct · MAJOR = blocks sign-off · MODERATE = should fix before delivery · MINOR = cosmetic, fix if time allows